Question 1: Choosing an SVM kernel

Kernel	Degree	C	γ	Mean CV Error	SD CV Error
RBF	—	1	0.01	0.2726	0.0371
Linear	—	0.01	—	0.2856	0.0235
Linear	—	10	—	0.2877	0.0551
Linear	—	0.1	—	0.2898	0.0584
Linear	—	1	—	0.2920	0.0620
RBF	—	1	0.1	0.2986	0.0350
Polynomial	3	1	scale	0.2987	0.0244
RBF	—	10	0.1	0.3073	0.0409
Polynomial	2	1	scale	0.3138	0.0227
Polynomial	2	10	scale	0.3161	0.0227
Polynomial	4	1	scale	0.3181	0.0389
Polynomial	3	10	scale	0.3182	0.0246
Polynomial	4	10	scale	0.3225	0.0213
Polynomial	3	0.1	scale	0.3290	0.0286
Polynomial	2	0.1	scale	0.3334	0.0166
Polynomial	4	0.1	scale	0.3334	0.0262
Polynomial	3	0.01	scale	0.3399	0.0115
Polynomial	4	0.01	scale	0.3421	0.0141
RBF	—	0.1	0.01	0.3463	0.0020
Polynomial	2	0.01	scale	0.3463	0.0020

We compared 20 SVM candidate settings across linear, polynomial (degrees 2, 3, and 4), and RBF kernels using the same stratified five-fold cross-validation. Predictors were standardized using only the training observations within each fold. The polynomial kernels used `gamma="scale"` and `coef0=0`.

The RBF kernel with C = 1 and γ = 0.01 achieved the lowest mean CV misclassification rate of 0.2726. The best linear model had a slightly higher error rate of 0.2856, while the polynomial kernels performed worse. Therefore, I prefer the RBF model with C = 1 and γ = 0.01 among the 20 candidates, although its improvement over the linear model is relatively small.

Question 2: Write and use a short SVM analysis skill

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name: svm-classification

description: Compare SVM classifiers (Linear, Polynomial, RBF) with tuned hyperparameters and 5-fold CV, then produce a one-page report.

SVM Classification

Find a good SVM classifier. Keep the process clear and the explanation short.

1. Check the data

- Find target and predictors; check types, sample size, missing values, number of classes.
- Show count and % per class. If a class has < 5 samples, explain why 5-fold CV fails.

2. Prepare the data

- Encode categoricals, handle missing values, standardize numerics.
- Use a Pipeline so preprocessing is fit only on each training fold (no leakage).

3. Kernels and grid

- Linear (baseline): C. Polynomial: C, degree, gamma. RBF: C, gamma.
- Default grid: C = [0.1, 1, 10], degree = [2, 3], gamma = ['scale', 0.01, 0.1, 1].

4. Compare fairly

- Same shuffled, stratified 5-fold CV for all models (seed = 42).
- Report Mean CV Error and SD across folds; pick the lowest mean.
- If classes are very unbalanced, note error rate may mislead.

5. Save the results

- CSV: Kernel, Degree, C, Gamma, Mean CV Error, SD CV Error; all rows, sorted low to high.
- PNG: dot plot with error bars (`ax.errorbar`, fmt="o", capsize=4), never bars.
  Groups in order Linear (tab:blue), RBF (tab:orange), Polynomial (tab:green),
  each sorted by Mean CV Error, 2-slot gap between groups, figsize=(17, 7).
  Per-point labels below axis, rotated 55, fontsize 8: Linear "C=..",
  RBF "C=..\nγ=..", Poly "d=..\nC=..". Bold group names as x ticks (pad=110).
  Title "SVM Kernel and Hyperparameter Comparison", y-label
  "Mean CV Misclassification Rate", light y-grid, bottom margin 0.35.

- PDF report: STRICTLY ONE PAGE. 11pt body, 0.6" margins, figure <= 2.2" tall,
  table = best 2 rows per kernel only, each section <= 3 sentences.
  After building, count pages; if > 1, shrink figure/font and rebuild until 1 page.
- Save the analysis code so it can be rerun.

6. Explain the result

- Best kernel, parameters, Mean CV Error, SD; compare briefly with Linear.
- Note limitations; don't overstate tiny differences. Short, plain English.

SVM Classification of Coronary Heart Disease (SAheart)

1. Data and Method

462 observations, 9 predictors; response chd (302 no, 65.4%; 160 yes, 34.6%), no missing values. famhist was dummy-coded and numeric predictors standardized inside a Pipeline. Linear, Polynomial, and RBF SVMs were tuned over C = {0.1, 1, 10}, degree = {2, 3}, gamma = {scale, 0.01, 0.1, 1} using the same stratified 5-fold CV (seed = 42).

2. Results

Kernel	Degree	C	Gamma	Mean CV Error	SD
linear	-	0.1	-	0.2877	0.0452
linear	-	10	-	0.2877	0.0492
rbf	-	10	0.01	0.2618	0.0290
rbf	-	1	0.1	0.2813	0.0344
poly	2	1	scale	0.2770	0.0189
poly	2	10	scale	0.2856	0.0254

Table 1. Best two settings per kernel (highlighted = selected).

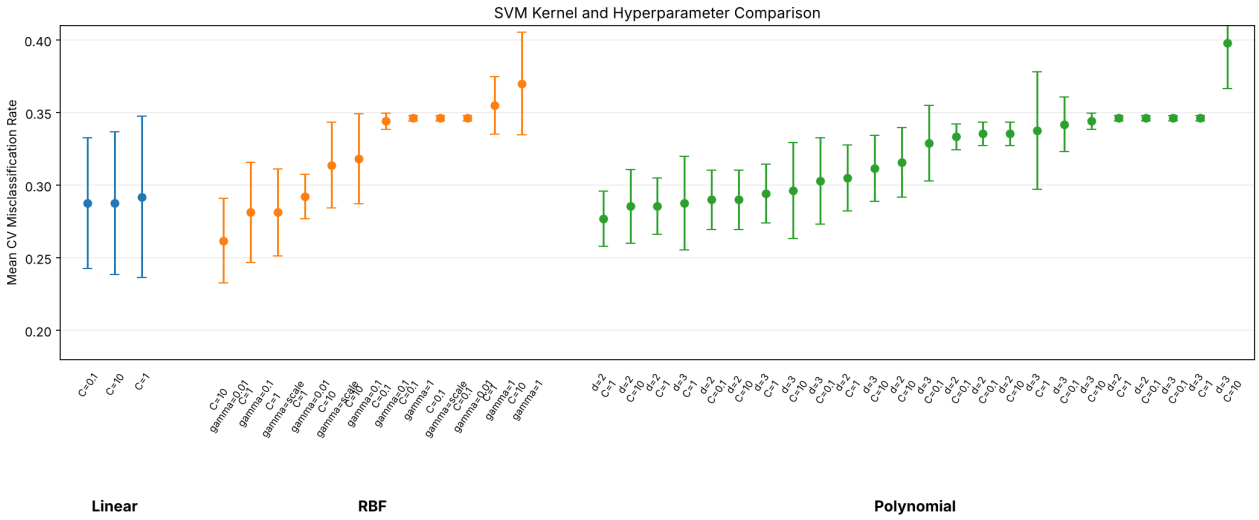


Figure 1. Mean CV error +/- 1 SD for every setting, grouped by kernel.

3. Selected Model

RBF SVM with C = 10, gamma = 0.01 had the lowest error: 0.2618 (SD 0.0290). The best Linear SVM (C = 0.1) reached 0.2877 (SD 0.0452), about 2.6 points worse. The gap is smaller than the fold-to-fold SD, so the boundary is close to linear.

4. Limitations

Differences between top models are within one SD and may not hold on new data. The grid is coarse and no separate test set was used, so CV error is slightly optimistic for the chosen model. With 35% positives, error rate hides sensitivity to heart-disease cases.